

Banking Customer Portfolio & Lending Performance Analysis

Executive Summary

This report presents a structured analysis of a banking dataset comprising 3,000 customers across demographics, financial products, and lending activity. The goal was straightforward: look at the data, find what actually matters, and translate those findings into clear takeaways that a business can act on.

A few things stood out immediately. The bank's loan portfolio is substantial — total loans across retail and business lending comfortably exceed total deposits — which signals a healthy credit operation but also one that needs to be managed carefully. Medium-income customers carry the most weight across nearly every product category. And European customers, while dominant in volume purely because of their share of the customer base, show similar per-customer engagement to other nationalities, which tells a different story than headline numbers alone.

The analysis also surfaced some quieter signals worth paying attention to: over three-quarters of customers own property, the average customer is in their early fifties, and the Jade loyalty tier — the bank's entry-level segment — holds the largest share of deposits, not Platinum. Each of these points has real implications for retention, cross-selling, and product design.

Project Context & Methodology

What This Project Set Out to Do

Financial institutions collect enormous amounts of customer data, but raw data on its own doesn't make decisions — analysis does. This project aimed to extract meaningful business intelligence from a retail and commercial banking dataset, with a focus on understanding customer behavior, evaluating portfolio composition, and identifying areas where the bank could grow or protect value.

How the Analysis Was Conducted

The dataset was first explored using Python in a Jupyter Notebook to understand its structure, check for anomalies, and calculate summary statistics. Power BI was then used to build an interactive dashboard for visual analysis across segments. DAX measures were written to calculate KPIs and enable dynamic comparisons. The final report synthesizes both layers of analysis.

The dataset includes 25 variables across three broad categories:

- Customer profile — age, gender, nationality, occupation, loyalty tier, income band
- Financial products — deposits, savings accounts, checking accounts, credit card balances, superannuation
- Lending & risk — bank loans, business lending, risk weighting, property ownership

Key Performance Indicators

The table below summarises the top-level metrics derived from the dataset. These numbers form the foundation for the analysis that follows.

Metric	Value	Context
Total Customers	3,000	Retail and commercial clients
Total Bank Deposits	₹2.01 Billion	Primary funding source
Total Bank Loans	₹1.77 Billion	Retail lending portfolio
Total Business Lending	₹2.60 Billion	Largest single product segment
Savings Accounts	₹698.7 Million	Avg ₹232,908 per customer
Checking Accounts	₹963.3 Million	Avg ₹321,093 per customer
Loan-to-Deposit Ratio	0.88	Healthy; below 1.0 threshold
Avg Customer Age	51 years	Range: 17–85
Property Ownership Rate	76.5%	2,295 of 3,000 customers

Key Findings & Analysis

1. Business Lending Is the Dominant Product — and a Concentration Risk

At ₹2.60 billion, business lending is larger than the entire retail loan book and nearly equals total deposits. On the surface this looks like a strong commercial banking operation, and it is. But it also means the bank's overall financial health is heavily tied to how business clients perform.

The average business lending per customer is ₹866,760 — nearly three times the average retail loan of ₹591,386. That gap matters: a deterioration in even a small cohort of large commercial borrowers would have an outsized effect on the portfolio. The recommendation here is not to pull back from business lending, but to ensure that relationship management and credit monitoring are proportionally resourced for this segment.

2. The Loan-to-Deposit Ratio Tells a Reassuring Story — For Now

The loan-to-deposit ratio sits at 0.88, meaning the bank is lending out roughly 88 cents for every dollar it holds in deposits. This is within a comfortable range; regulators typically flag ratios above 1.0 as a potential liquidity concern. The bank is not over-extended.

However, when business lending is factored in, total credit exposure significantly exceeds deposit holdings. This is not unusual for a bank with a strong commercial book, but it underscores the importance of maintaining stable deposit growth alongside lending expansion.

3. Medium-Income Customers Are the Real Engine of the Business

Income was segmented into three bands for this analysis: below ₹100,000 (lower income), ₹100,000–₹250,000 (middle income), and above ₹250,000 (high income). The middle income band accounts for 1,262 customers — 42% of the base — and consistently shows the broadest product engagement across deposits, loans, and savings.

High-income customers (711 in total) show the largest average balances — their average deposit of ₹919,841 is nearly 37% above the overall mean — but they are fewer in number. Lower-income customers (1,027) tend to hold smaller balances across all products. The medium-income segment is not the most profitable per customer, but it is the most stable and the widest. It deserves dedicated retention programs, not just acquisition campaigns.

4. The Jade Loyalty Tier Deserves More Attention Than It Gets

Loyalty classification runs from Jade (entry-level) through Silver, Gold, and Platinum. Counterintuitively, Jade customers — the largest group at 1,331 — hold the highest average deposits (₹696,577) and business lending (₹877,877) among all tiers. Platinum customers, despite being the most premium segment, show lower averages on both metrics.

This is a meaningful finding. It suggests either that loyalty tier assignment does not cleanly track product value, or that a significant number of high-value customers are sitting in the Jade tier without having been meaningfully engaged or upgraded. Either way, there is likely a retention and upgrade opportunity buried in this segment that is currently going unaddressed.

5. European Customers Lead in Volume, But Per-Customer Averages Are Comparable Across Groups

European customers make up 43.6% of the base (1,309 customers) and therefore account for the largest share of total deposits (₹873.9 million). But their average deposit per customer (₹667,659) is actually slightly below the averages for Asian (₹683,438) and African (₹687,868) customers.

This matters for strategy. The bank's exposure to European customers is high in absolute terms — which creates some geographic concentration risk — but the individual-level engagement across nationalities is broadly similar. That is a positive sign for portfolio resilience. It also suggests that marketing efforts targeting minority nationality segments could yield meaningful volume gains without needing outsized investment.

6. Risk Profile Is Skewed Toward the Safe End — With a Notable Tail

Risk weighting runs on a scale of 1 to 5. About 68.6% of customers fall in the low-to-moderate risk category (ratings 1–2). Only 16.1% sit in the high-risk bands (4–5). This is a healthy distribution for a retail-focused book.

That said, 482 customers in the high-risk category is not a small number in absolute terms. And because these customers tend to hold larger-than-average loan balances — a pattern that often correlates with higher risk scores — their collective exposure should be tracked closely. A proactive outreach program for risk-band 4 customers, before they migrate to band 5, would be worth building.

7. The Customer Base Is Older Than Average — and That Shapes Product Needs

The mean customer age is 51, with the full range spanning from 17 to 85. This age profile has a few practical implications. First, superannuation savings and property ownership rates (76.5%) suggest a customer base that is in or approaching wealth consolidation mode rather than wealth accumulation. These customers are less likely to respond to new credit products and more likely to value wealth management, estate planning, and premium service offerings.

Second, a 51-year average means the bank's natural attrition rate — customers aging out or passing on — will increase over the next decade. Acquiring younger customers now, even if their balances are smaller, is an investment in the long-term health of the deposit base.

Strategic Recommendations

The following recommendations are drawn directly from the data findings above. They are ordered by the degree to which they address the bank's most material risks and opportunities.

Priority Recommendations

- Review Jade tier classification criteria.

The data strongly suggests that a meaningful number of high-value customers are not being identified or treated as such. A re-evaluation of how loyalty tiers are assigned — and a targeted upgrade campaign — could improve retention and share-of-wallet at relatively low cost.

- Build a proactive risk monitoring program for bands 4–5.

With 482 high-risk customers holding disproportionately large loan balances, early intervention is far cheaper than late-stage recovery. A quarterly review cadence for this cohort, with dedicated relationship manager outreach, is a sensible first step.

- Develop a medium-income retention strategy.

This segment drives the most volume and the most stable recurring activity. It is also the most vulnerable to churn if competitors offer better rates or digital experiences. Personalised product bundles, loyalty perks, and proactive re-engagement at key life milestones (property purchase, retirement planning) would all serve this segment well.

Growth Opportunities

- Target younger customer acquisition now.

An average customer age of 51 is a manageable situation today, but it becomes a structural problem in ten years. Graduate banking packages, digital-first product design, and partnerships with employers or universities could help shift the demographic mix.

- Explore premium wealth offerings for property-owning, high-income customers.

With 711 high-income customers, an average income of ₹171,305 across the full base, and a 76.5% property ownership rate, there is a credible segment for premium services: mortgage refinancing, investment accounts, and wealth management advice. These customers currently likely hold more assets outside the bank than within it.

- Test occupation-based product campaigns.

Occupation shows meaningful variation in product usage. The top occupations in the dataset include engineers, HR professionals, academic staff, and coordinators — each with different financial needs. Targeted campaigns by professional segment are more likely to convert than generic outreach.

Limitations & Caveats

Every analysis has constraints, and it is worth being upfront about them here.

- The dataset does not include time-series data, so it is not possible to assess trends over time — growth, seasonality, or whether the portfolio has improved or deteriorated.
- Risk weighting is a derived field, and without knowing the exact scoring methodology, the segments defined in this report (low, moderate, high) are approximate.
- Occupation data is highly granular (many distinct job titles) and was not fully normalised for this analysis. A more structured occupational taxonomy would enable cleaner segmentation.
- The dataset does not include transaction-level data, so behavioral insights (frequency of use, channel preference, dormancy) are not available.
- Currency and geography assumptions are implied from the data structure; without explicit documentation, some regional benchmarks may not apply directly.

Conclusion

This project began as an exercise in working with banking data, and it ended up surfacing some genuinely useful business intelligence. The headline finding — that business lending is the bank's growth driver — is not surprising for a commercial bank, but the detail matters: the concentration in that segment creates risk that deserves active management.

What was more interesting was the loyalty tier anomaly. High-value customers sitting in the entry-level Jade tier is the kind of finding that only shows up when you look at the data carefully, and it has real commercial value. So does the age story — not alarming now, but something that should be shaping decisions about product design and customer acquisition today, not in five years.

Good data analysis is not just about calculating aggregates. It is about knowing which numbers to put side by side and asking why the pattern exists. This report tried to do that, and the recommendations it surfaces are grounded in what the data actually shows, not what might be expected.

Prepared as part of a data analytics portfolio project. All figures derived from the provided dataset of 3,000 banking customer records.